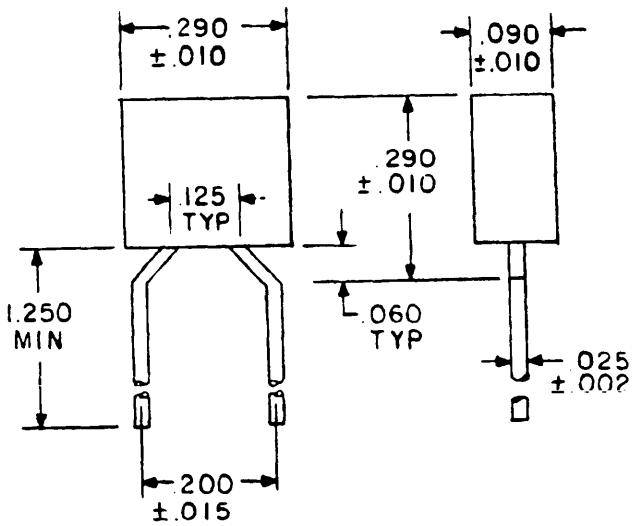
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), ESTABLISHED RELIABILITY, STYLE CKR03

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification sheet and the latest issue of MIL-C-39014.



INCHES	MM
.010	0.25
.015	0.38
.060	1.52
.125	3.18
.200	5.08
.290	7.37
1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Lead length may be a minimum of one inch for use in tape and reel automatic insertion equipment, when specified in the ordering data.

FIGURE 1. Dimensions and configuration.

Ⓐ denotes changes

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Epoxy coated.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55°C to +125°C.

Voltage rating: See table I.

Radiographic inspection: Not applicable.

Dissipation factor: Not more than 2.5 percent.

Dielectric withstanding voltage: Body insulation shall be able to withstand 1300 V dc between leads and case.

Insulation resistance:

At 25°C: Not less than 100,000 megohms or 1,000 megohm-microfarads whichever is less.

At 125°C: Not less than 10,000 megohms or 100 megohm-microfarads whichever is less.

Seal: Not applicable.

Voltage-temperature limits: In accordance with MIL-C-39014.

Capacitance change with reference to 25°C	
Steps A through D of table V Bias = 0 volts	Steps E through G of table V Bias = rated voltage
±15 percent	+15, -25 percent as applicable (see table I)

Immersion: IR not less than 50 percent of initial 25°C requirement.

Salt spray (corrosion): Not applicable.

Ⓐ Resistance to soldering heat: ΔC +15, -5 percent of the initial 25°C measurement.

Life:

Rated conditions: 100 percent of dc rated voltage at 125°C.
0 through 3000 hours:
IR: At 25°C, not less than 50 percent of initial 25°C requirement.
At 125°C, not less than 50 percent of initial 125°C requirement.
ΔC: ±20 percent from initial measured value.
DF: Not more than 2.5 percent.

4,000 through 32,000 hours:
IR: At 25°C, not less than 15 percent of initial 25°C requirement.
At 125°C, not less than 15 percent of initial 125°C requirement.
ΔC: ±20 percent from initial measured value.
DF: Not greater than 2.5 percent.

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Accelerated condition: 200 percent of dc rated voltage at 125°C.
0 and 250 hours:
IR: At 25°C, not less than 50 percent of initial 25°C requirement.
At 125°C, not less than 50 percent of initial 125°C requirement.
ΔC: ±20 percent from initial measured value.
DF: Not more than 2.5 percent.

1,000 2,000 and 4,000 hours:
IR: At 25°C, not less than 15 percent of initial 25°C requirement.
At 125°C, not less than 15 percent of initial 125°C requirement.
ΔC: ±20 percent from initial measured value.
DF: Not more than 2.5 percent.

Part number: M39014/24-(dash number from table I).

Marking: In accordance with MIL-C-39014.

TABLE I. Dash number and electrical characteristics.

Dash number				Capacitance	Capacitance tolerance	DC rated voltage
Failure rate level (%/1,000 hours)						
1.0 (M)	0.1 (P)	0.01 (R)	0.001 (S)	pF	percent	volts
0001	0101	0201	0301	560,000	10	50
0002	0102	0202	0302	680,000	10	50
0003	0103	0203	0303	820,000	10	50
0004	0104	0204	0304	1,000,000	10	50

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - MI, AR
Air Force - 17, 85
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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